

Partial Differential Equations

Fall 2025, Math 475 Course Notes

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September 23, 2025

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Acknowledgments

I would like to extend my deepest thanks and appreciation to the following people, without whose support this note would not have been possible:

[**Adam Oberman**], *Professor, McGill University*

I would like to express my deepest gratitude to Professor Oberman, the instructor for this course, whose guidance and expertise were invaluable throughout the development of my notes.

Despite all efforts, there may still be some typos, unclear explanations, etc. If you find potential mistakes, or any suggestions regarding concepts or formats, etc., feel free to reach out to the author at *zhangjohnson729@gmail.com*.

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First-Order Linear PDEs

In this section, we will dive into some basic and elementary structures of partial differential equations. Throughout the section, consider $u(x_1, \dots, x_n)$ as a multivariate function, $u_{x_1}, \dots, u_{x_n}$ as its first order partial derivatives, and $u_{x_i x_j} \dots$ as second order partial derivatives and so on.

1.1 Constant Coefficient Linear PDEs

In this part, let $u(x, y) : U \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$, $a, b \in \mathbb{R}$ not both zero, and we are interested in the PDE of the form

$$\boxed{au_x + bu_y = 0} \quad (1.1)$$

A very intuitive example is $u(x, y) = bx - ay$, where $u_x = b, u_y = -a$ and $au_x + bu_y = ab - ab = 0$. But in general how can we solve it? We will give two methods.

1.1.1 Geometric Method Via Directional Derivative

When we have a multivariate function (in our example it suffices to illustrate with 2 variables) $z = u(x, y)$, the two partial derivatives are defined by

Definition 1. Let $z = u(x, y)$ be a multivariate function, then the partial derivatives u_x, u_y at (x_0, y_0) are defined as

$$u_x := \lim_{\Delta x \rightarrow 0} \frac{u(x_0 + \Delta x, y_0) - u(x_0, y_0)}{\Delta x} \quad (1.2)$$

$$u_y := \lim_{\Delta y \rightarrow 0} \frac{u(x_0, y_0 + \Delta y) - u(x_0, y_0)}{\Delta y}. \quad (1.3)$$

Just like normal derivative with one variable, the partial derivatives u_x, u_y are the rate of change along the x, y axis respectively. We may further extend this idea to any direction, say the “partial derivative” or the “rate of change” along the line $y = x$, $y = 2x - 3$ (a linear combination of x, y coordinates) etc. That’s where we introduce directional derivatives. So we may then define the directional derivative along a vector $\mathbf{v} = (a, b)$:

Definition 2. The directional derivative of $u(x, y)$ at (x_0, y_0) along $\mathbf{v} = (a, b)$ is given by

$$D_{\mathbf{v}}u(x_0, y_0) = \lim_{h \rightarrow 0} \frac{u(x_0 + ha, y_0 + hb) - u(x_0, y_0)}{h} \quad (1.4)$$

and we can easily see that u_x, u_y is just the special case when $\mathbf{v} = (1, 0), (0, 1)$ respectively. Below is a figure to illustrate directional derivative geometrically:

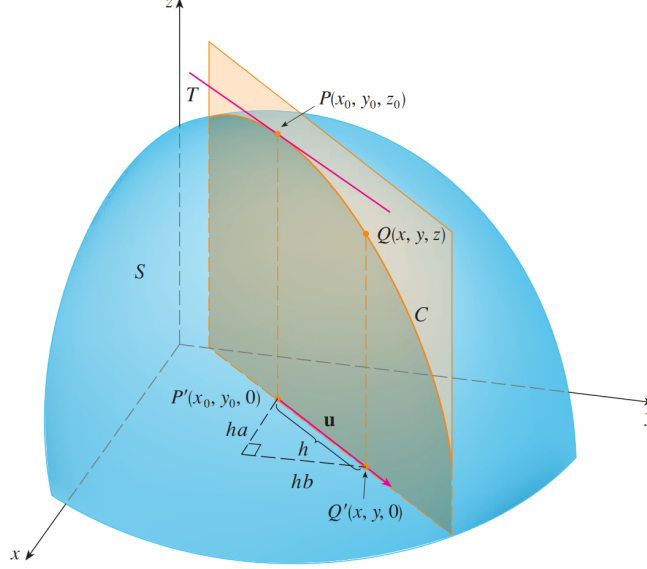


Figure 1: The directional derivative at (x_0, y_0, z_0) of the function $z = u(x, y)$. It can be viewed as the slope of tangent line of the curve obtained by slicing the function with a vertical plane that passes through the direction of the directional derivative. (Credit: *James-Stewart Calculus, early transcendentals, Eighth Edition, Page 946*)

Theorem 1. The directional derivative of $u(x, y)$ along $\mathbf{v} = (a, b)$ can be computed by

$$D_{\mathbf{v}}u(x, y) = u_x(x, y) \cdot a + u_y(x, y) \cdot b \quad (1.5)$$

The proof of this theorem can be done by using Chain rule and hence the readers are encouraged to try it themselves.

After knowing some directional derivatives, we go back to the equation $a \cdot u_x + b \cdot u_y = 0$, we realized this means the directional derivative of $u(x, y)$ along $\mathbf{v} = (a, b)$ is equal to zero, meaning $u(x, y)$ is constant along $\mathbf{v}$. The line equation of $\mathbf{v}$ can be expressed by $bx - ay = 0$, and the set of lines parallel to $\mathbf{v}$ has the general form of $bx - ay = C$, where $C \in \mathbb{R}$. They are called the *characteristic lines*, and on each of those lines $D_{\mathbf{v}}u(x, y)$ is a constant and the entire plane $\mathbb{R}^2$ is generated by the set of all characteristic lines. So we see that in this case $u(x, y)$ only depends on $bx - ay$, i.e which characteristic line it belongs to, and hence the general solution to $au_x + bu_y = 0$ is

$$\boxed{u(x, y) = f(bx - ay)} \quad (1.6)$$

where f is a function of one variable, and the exact f may be obtained once extra conditions are given. To verify the correctness of (1.6), we two partial derivatives:

$$u_x = bf'(bx - ay), u_y = -af'(bx - ay) \quad (1.7)$$

and hence $au_x + bu_y = abf'(bx - ay) - abf'(bx - ay) = 0$.

Example 1. Solve the PDE $4u_x - 3u_y = 0$ with auxiliary condition $u(0, y) = y^3$.

Solution: It is first easy to see that $u(x, y) = f(-3x - 4y)$ is the general solution, where $-3x - 4y = C$ is the characteristic line. Then plugging in the auxiliary condition, $y^3 = u(0, y) = f(-4y)$ we get $f(-4y) = y^3$ and by a change of variable we see that $f(t) = -t^3/64$. So we have $u(x, y) = f(-3x - 4y) = (-3x - 4y)^3/64$.

Example 2. Solve the PDE $xu_x + yu_y = 0$.

We notice that the directional derivatives along (x, y) is constant zero, and we have

$$\frac{dy}{dx} = \frac{y}{x} \quad (1.8)$$

which brings us to a separable differential equation and we have

$$\int \frac{1}{y} dy = \int \frac{1}{x} dx + C \quad (1.9)$$

and we have $\ln |y| - \ln |x| = C$, which is the characteristic curve of the PDE, and given any function g defined through $g(\ln |y| - \ln |x|)$ will be a solution.

1.1.2 Change of Variable Method

Now we present a different method using change of variables. Imagine using the direction of $\mathbf{v} = (a, b)$ as our new coordinates x' and $\mathbf{v}^\perp$ as y' . Then we can verify that

$$x' = ax + by \quad y' = bx - ay \quad (1.10)$$

and hence we have the relation that

$$x = \frac{ax' + by'}{a^2 + b^2} \quad y = \frac{bx' - ay'}{a^2 + b^2} \quad (1.11)$$

and by Chain rule, we get

$$u_x = \frac{\partial u}{\partial x'} \cdot \frac{\partial x'}{\partial x} + \frac{\partial u}{\partial y'} \cdot \frac{\partial y'}{\partial x} = au_{x'} + bu_{y'} \quad (1.12)$$

$$u_y = \frac{\partial u}{\partial x'} \cdot \frac{\partial x'}{\partial y} + \frac{\partial u}{\partial y'} \cdot \frac{\partial y'}{\partial y} = bu_{x'} - au_{y'} \quad (1.13)$$

so we have

$$au_x + bu_y = a(au_{x'} + bu_{y'}) + b(bu_{x'} - au_{y'}) = (a^2 + b^2)u_{x'} \quad (1.14)$$

and we obtain a new equation $(a^2 + b^2)u_{x'} = 0$, which indicates the solution $u(x, y)$ is independent of x' and hence we have $u(x', y') = f(y') = f(bx - ay)$. Note that this is the exact same answer we get using the directional derivative method.

Then, we are interested in a more general form

$$\boxed{au_x + bu_y = f(x, y)} \quad (1.15)$$

for a given function $f(x, y)$. If f reduces to zero then it is the case in section 1.1.1.

Theorem 2. The solution to (1.13) is given by

$$u(x, y) = \frac{1}{\sqrt{a^2 + b^2}} \int_L f ds + g(bx - ay) \quad (1.16)$$

where L is the characteristic line segment from y axis to (x, y) .

Proof. We will again use the change of variable method, using the change of variables discussed in (1.8), and the relations stated in (1.9);(1.10);(1.11), we have

$$a(au_{x'} + bu_{y'}) + b(bu_{x'} - au_{y'}) = f \left(\frac{ax' + by'}{a^2 + b^2}, \frac{bx' - ay'}{a^2 + b^2} \right) \quad (1.17)$$

which simplifies to

$$u_{x'} = \frac{1}{a^2 + b^2} \cdot f \left(\frac{ax' + by'}{a^2 + b^2}, \frac{bx' - ay'}{a^2 + b^2} \right) \quad (1.18)$$

hence integrate 1.16 with respect to x' gives

$$u(x', y') = \int_{x'} \frac{1}{a^2 + b^2} \cdot f \left(\frac{ar + by'}{a^2 + b^2}, \frac{br - ay'}{a^2 + b^2} \right) dr + g(y') \quad (1.19)$$

and we now use the original coordinates and we have

$$u(x, y) = \int_{ax+by} \frac{1}{a^2 + b^2} \cdot f \left(\frac{ar + b(bx - ay)}{a^2 + b^2}, \frac{br - a(bx - ay)}{a^2 + b^2} \right) dr + g(bx - ay) \quad (1.20)$$

by a simple change of variable $s = \frac{r}{\sqrt{a^2 + b^2}}$, we have

$$u(x, y) = \frac{1}{\sqrt{a^2 + b^2}} \int_{\frac{ax+by}{\sqrt{a^2+b^2}}} f \left(\frac{b^2x + a(\sqrt{a^2 + b^2}s - by)}{a^2 + b^2}, \frac{a^2y + b(\sqrt{a^2 + b^2}s - ax)}{a^2 + b^2} \right) ds + g(bx - ay) \quad (1.21)$$

and denoted by

$$u(x, y) = \frac{1}{\sqrt{a^2 + b^2}} \int_L f ds + g(bx - ay) \quad (1.22)$$

and as we can see L is the line segment from y axis to the point (x, y) . ■

Example: Solve the PDE $u_x + u_y = 1$.

Solution: We follow the result of theorem 2, we have $f(x, y) = 1, a = b = 1$, then we plug these values into (1.20), we have

$$\begin{aligned} u(x, y) &= \frac{1}{\sqrt{2}} \int_{\frac{x+y}{\sqrt{2}}} ds + g(x - y) \\ &= \frac{x + y}{2} + g(x - y) \end{aligned}$$

and we get

$$u_x = \frac{1}{2} + g'(x - y) \quad u_y = \frac{1}{2} - g'(x - y) \quad (1.23)$$

where $g(x - y)$ is an arbitrary single variable function that depends through $x - y$, and notice that $u_x + u_y = 1$ which means the solution is valid.

Example: Solve the PDE $u_x + u_y = xy$.

Solution: In this example, we have $f(x, y) = xy, a = b = 1$ and (1.20) gives us

$$u(x, y) = \frac{1}{\sqrt{2}} \int_{\frac{x+y}{\sqrt{2}}} \left(\frac{x + \sqrt{2}s - y}{2} \right) \cdot \left(\frac{y + \sqrt{2}s - x}{2} \right) ds + g(x - y) \quad (1.24)$$

$$= \frac{1}{\sqrt{2}} \int_{\frac{x+y}{\sqrt{2}}} \frac{2s^2 - (x - y)^2}{4} ds + g(x, y) \quad (1.25)$$

$$= \left[\frac{2s^3 - 3s(x - y)^2}{12} \right] \Big|_{s=\frac{x+y}{\sqrt{2}}} + g(x - y) \quad (1.26)$$

$$= -\frac{(x + y)(x^2 + y^2 - 4xy)}{12} + g(x - y) \quad (1.27)$$

hence we get the general solution to the original PDE:

$$u(x, y) = -\frac{(x + y)(x^2 + y^2 - 4xy)}{12} + g(x - y) \quad (1.28)$$

where $g(x - y)$ is an arbitrary single valued function which depends through $x - y$. To verify, note that we have

$$\begin{aligned} u_x &= -\frac{1}{12} \left(x^2 + y^2 - 4xy + (x + y)(2x - 4y) \right) + g'(x - y) \\ &= \frac{1}{4} \left(-x^2 + y^2 + 2xy \right) + g'(x - y) \end{aligned} \quad (1.29)$$

and similarly

$$\begin{aligned} u_y &= -\frac{1}{12} \left(x^2 + y^2 - 4xy + (x + y)(2y - 4x) \right) - g'(x - y) \\ &= \frac{1}{4} \left(x^2 - y^2 + 2xy \right) - g'(x - y) \end{aligned} \quad (1.30)$$

where

$$\begin{aligned} u_x + u_y &= \frac{1}{4} \left(-x^2 + y^2 + 2xy + x^2 - y^2 + 2xy \right) + g'(x - y) - g'(x - y) \\ &= xy. \end{aligned} \quad (1.31)$$

1.2 Nonhomogeneous Problem and Method of Characteristics

We now generalize the idea of transport equation into $\mathbb{R}^n$. Suppose $x, b \in \mathbb{R}^n, t \in (0, \infty)$ and Du is the gradient of u with respect to x_i , then consider the problem

$$\begin{cases} u_t + b \cdot Du = f & \text{in } \mathbb{R}^n \times (0, \infty) \\ u = g & \text{on } \mathbb{R}^n \times \{t = 0\} \end{cases} \quad (1.32)$$

To solve this, we inspired by the homogeneous problem, suppose we set $z(s) := u(x + sb, t + s)$ for $s \in \mathbb{R}$, then differentiating z will yield

$$\dot{z}(s) = b \cdot Du(x + sb, t + s) + u_t(x + sb, t + s) = f(x + sb, t + s) \quad (1.33)$$

then we notice that $u(x, t) = u(x + 0b, t + 0) = z(0)$ and $u(x - tb, t - s) = u(x - tb, 0) = g(x - tb) = z(-t)$, so we have

$$u(x, t) - g(x - tb) = z(0) - z(-t) = \int_{-t}^0 \dot{z}(s) ds \quad (1.34)$$

$$= \int_{-t}^0 f(x + sb, t + s) ds \quad (1.35)$$

$$= \int_0^t f(x + (s - t)b, s) ds \quad (1.36)$$

hence, we have the general solution given by

$$u(x, t) = g(x - tb) + \int_0^t f(x + (s - t)b, s) ds \quad (1.37)$$

Example 3. Solve the PDE $u_t + u_x = 1$ with $u(x, 0) = x^2$.

In this case observe that $f \equiv 1$, $g = x^2$, $Du = u_x$, $b = 1$, so by the formula we proposed, we have

$$u(x, t) = (x - t)^2 + \int_0^t 1 ds = x^2 - 2xt + t^2 + t. \quad (1.38)$$

It is easy to see that the function above solves the PDE.

We now study a more general form: $F(Du, u, x) = 0$ in U subject to the boundary condition $u = g$ on Γ where $\Gamma \subseteq \partial U$, $g : \Gamma \rightarrow \mathbb{R}$, also $F, g \in C^\infty$. So what is the *method of characteristics*? Well we pick $x \in U$, and then another $x_0 \in \Gamma$. Since we already know on Γ , $u = g$ so we are able to compute the value at x_0 . Then we try to connect x_0, x by some nice curve u that solves the PDE.

Suppose such a curve which is described parametrically by $\mathbf{x}(s) = (x^1(s), \dots, x^n(s))$ where $s \in \mathbb{R}$ is our parameter. We assume $u \in C^2$ solves the PDE $F(Du, u, x) = 0$ and let $z(s) = u(\mathbf{x}(s))$, as well as $\mathbf{p}(s) := Du(\mathbf{x}(s))$ where $p^i(s) = u_{x_i}(\mathbf{x}(s))$. In our set up, z gives the values along the curve, $\mathbf{p}$ gives the values of the gradient Du . Note that

$$\dot{p}^i(s) := \frac{d}{ds} p^i(s) = \frac{d}{ds} \sum_{j=1}^n u_{x_i x_j}(\mathbf{x}(s)) \dot{x}^j(s) \quad (1.39)$$

also we differentiate $F(Du, u, x) = 0$ with respect to x^i :

$$\sum_{j=1}^n \frac{\partial F}{\partial p_j}(Du, u, x) u_{x_j x_i} + \frac{\partial F}{\partial z}(Du, u, x) u_{x_i} + \frac{\partial F}{\partial x_i}(Du, u, x) = 0 \quad (1.40)$$

set

$$\dot{x}^j(s) = \frac{\partial F}{\partial p_j}(\mathbf{p}(s), z(s), \mathbf{x}(s)) \quad (1.41)$$

then we evaluate (1.40) at $x = \mathbf{x}(s)$, we have

$$\sum_{j=1}^n \frac{\partial F}{\partial p_j}(\mathbf{p}(s), z(s), \mathbf{x}(s)) u_{x_i x_j}(\mathbf{x}(s)) + \frac{\partial F}{\partial z}(\mathbf{p}(s), z(s), \mathbf{x}(s)) p^i(s) + \frac{\partial F}{\partial x_i}(\mathbf{p}(s), z(s), \mathbf{x}(s)) = 0 \quad (1.42)$$

so now use the expression above and (1.41), substitute back into (1.39) will give us

$$\dot{p}^i(s) = -\frac{\partial F}{\partial x_i}(\mathbf{p}(s), z(s), \mathbf{x}(s)) - \frac{\partial F}{\partial z}(\mathbf{p}(s), z(s), \mathbf{x}(s)) p^i(s) \quad (1.43)$$

and finally differentiating $z = u(\mathbf{x}(s))$ will yield

$$\dot{z}(s) = \sum_{j=1}^n \frac{\partial u}{\partial x_j}(\mathbf{x}(s)) \dot{x}^j(s) = \sum_{j=1}^n p^j(s) \frac{\partial F}{\partial p_j}(\mathbf{p}(s), z(s), \mathbf{x}(s)). \quad (1.44)$$

Definition 3. We define the following systems of ODEs as the characteristic equations of the PDE $F(Du, u, x) = 0$:

$$\begin{cases} \dot{\mathbf{p}}(s) &= -D_x F(\mathbf{p}(s), z(s), \mathbf{x}(s)) - D_z F(\mathbf{p}(s), z(s), \mathbf{x}(s)) \mathbf{p}(s) \\ \dot{z}(s) &= D_p F(\mathbf{p}(s), z(s), \mathbf{x}(s)) \cdot \mathbf{p}(s) \\ \dot{\mathbf{x}}(s) &= D_p F(\mathbf{p}(s), z(s), \mathbf{x}(s)) \end{cases} \quad (1.45)$$

Expression (1.45) looks a little bit cursed, but let us consider a simple example:

$$a(x, y)u_x + b(x, y)u_y = c(x, y)u + d(x, y) \quad (1.46)$$

for function $u(x, y)$. Then the characteristic curve will be

$$\begin{cases} \dot{x}(s) = a(x(s), y(s)) \\ \dot{y}(s) = b(x(s), y(s)) \\ \dot{z}(s) = c(x(s), y(s))z + d(x(s), y(s)) \end{cases} \quad (1.47)$$

Example 4. Solve the PDE $-yu_x + xu_y = u$ with boundary condition $u(x, 0) = g(x)$.

It is easy to see that $\dot{x}(s) = -y, \dot{y}(s) = x, \dot{z}(s) = z$. We first solve the system of ODEs with x, y :

$$\begin{cases} \dot{x} = -y \\ \dot{y} = x \end{cases} \implies \begin{cases} x = c_2 \cos(s) - c_1 \sin(s) \\ y = c_1 \cos(s) + c_2 \sin(s) \end{cases} \quad (1.48)$$

Let $x(0) = x_0, y(0) = 0$, then $x(s) = x_0 \cos(s), y(s) = x_0 \sin(s)$. Furthermore solve for z we have $z(s) = c_3 e^s$. With initial condition, $z(0) = g(x_0)$ so we have $z(s) = g(x_0) e^s$. Now given (x, y) we need to solve for x_0 and s such that they passes through the characteristics, we solve $x = x_0 \cos(s), y = x_0 \sin(s)$ and hence $x_0^2 = x^2 + y^2, s = \arctan(y/x)$ and hence we obtain

$$u(x, y) = g(x_0) e^s = g(\sqrt{x^2 + y^2}) e^{\arctan(y/x)}. \quad (1.49)$$

Example 5. Solve the PDE $u_x + u_y + u = e^{x+2y}$ with boundary condition $u(x, 0) = 0$.

We first note that $\dot{x}(s) = 1, \dot{y}(s) = 1$ and $\dot{z}(s) = -1 + e^{x+2y}$, first solve the system of ODEs with x, y , we obtain

$$\begin{cases} \dot{x}(s) = 1 \\ \dot{y}(s) = 1 \end{cases} \implies \begin{cases} x(s) = s + c_1 \\ y(s) = s + c_2 \end{cases} \quad (1.50)$$

for some constant $c_1, c_2 \in \mathbb{R}$. We now plug in the initial condition, we know that $x(0) = x_0, y(0) = 0$ (we make it this way so the point lie on the “boundary” where we know $u = g$), hence we have

$$x(s) = s + x_0 \quad y(s) = s \quad (1.51)$$

with this information, we substitute x, y by s into z and have $\dot{z}(s) = -z(s) + e^{3s+x_0}$, which is an ODE that can be solved by using integrating factors. Note that

$$\mu(s) = \exp\left(\int 1 ds\right) \quad z(s) = \frac{1}{\mu(s)} \left(\int \mu(s)e^{3s+x_0} ds + C\right) \quad (1.52)$$

and we hence have

$$z(s) = \frac{1}{4}e^{x_0+3s} + \frac{C}{e^s}, C \in \mathbb{R} \quad (1.53)$$

with initial condition $z(0) = 0$, we have

$$z(s) = \frac{1}{4}e^{x_0+3s} - \frac{1}{4}e^{x_0-s} \quad (1.54)$$

we then use (1.51) to obtain x_0, s in terms of x, y , where $s = y, x_0 = x - y$, hence we have

$$u(x, y) = \frac{1}{4}(e^{x+2y} - e^{x-2y}). \quad (1.55)$$

The above two examples motivate us to further investigate (1.45) and derive a more specific formula for different type of PDEs:

Linear PDEs: The first case we consider is when $F(Du, u, x)$ is linear and homogeneous, which means it takes the form

$$b(x(s)) \cdot Du + c(x)u = 0 \quad (1.56)$$

and it suggests that we have $F(p, z, x) = b(x(s)) \cdot p(s) + c(x(s))z(s)$, where we have $D_p F = b(x)$ and then by (1.45)

$$\dot{x}(s) = D_p F = b(x(s)) \quad \dot{z}(s) = D_p F \cdot p(s) = b(x(s)) \cdot p(s) \quad (1.57)$$

also note that we have $F(p, x, z) = b(x) \cdot p(s) + c(x)z(s) = 0$ hence we further have the expression

$$\dot{z}(s) = -c(x(s))z(s) \quad (1.58)$$

so in summary we have

$$\begin{cases} \dot{x}(s) = b(x(s)) \\ \dot{z}(s) = -c(x(s))z(s) \end{cases} \quad (1.59)$$

as our parametrization and from this point we can easily see the parametrization we get in example 4 and 5.

Quasi-linear PDEs: We now consider the PDE of the form

$$F(Du, u, x) = b(x, u(x)) \cdot Du + c(x, u(x)) = 0 \quad (1.60)$$

and we now have the parametrization $F(p, x, z) = b(x, z) \cdot p + c(x, z) = 0$, and from here we have $D_p F = b(x(s), z(s))$ so $\dot{x}(s) = b(x(s), z(s))$, and $\dot{z}(s) = b(x(s), z(s)) \cdot p(s)$. We finally use the fact that $b(x, z) \cdot p + c(x, z) = 0$ to conclude $\dot{z}(s) = -c(x(s), z(s))$ so in summary we have

$$\begin{cases} \dot{x}(s) = b(x(s), z(s)) \\ \dot{z}(s) = -c(x(s), z(s)) \end{cases} \quad (1.61)$$

Example 6. Solve the PDE $u_x + u_y = u^2$ with boundary condition $u(x, 0) = g$.

We first see that $\dot{x} = 1, \dot{y} = 1$ so we have the solution

$$\begin{cases} x(s) = s + c_1 \\ y(s) = s + c_2 \end{cases} \quad (1.62)$$

with the boundary condition, we let x_0 be the point on g such that our parametrization yields $x(0) = x_0, y(0) = 0$ and hence we have

$$\begin{cases} x(s) = s + x_0 \\ y(s) = s \end{cases} \quad (1.63)$$

Now for z we have $\dot{z} = z^2$, which now turns into a separable ODE and we have

$$\int \frac{1}{z^2} dz = \int 1 ds + C \implies -\frac{1}{z} = s + C \quad (1.64)$$

take $s = 0$, and let $g(x_0) = z_0$, then we have $-\frac{1}{z_0} = C$ hence then we solve for z and we get

$$z(s) = \frac{z_0}{1 - sz_0} := \frac{g(x_0)}{1 - sg(x_0)} \quad (1.65)$$

Note that in (1.63) our parametrization will yield $s = y; x_0 = x - y$ so we substitute into (1.65) and we get

$$u(x, y) = \frac{g(x - y)}{1 - y \cdot g(x - y)} \quad (1.66)$$

as the solution to the PDE, where g is any function depend through $x - y$. But notice that the solution only makes sense only $1 - yg(x - y) \neq 0$.

Wave Equations

2.1 Interpretations and Derivations

Suppose $U \subseteq \mathbb{R}^n$ is open, and we define a function $u(x, t) : \bar{U} \times [0, \infty) \rightarrow \mathbb{R}$ where $x \in U$ and t is the time variable.

Definition 4. *The wave equation is defined by*

$$u_{tt} - \Delta u = 0 \quad (2.1)$$

where Δ is the Laplacian operator.

We first provide some physical interpretations to the wave equation. Suppose we have a smooth subregion $V \subseteq U$, and we consider the acceleration within V . The interpretation is easier for $n = 1, 2$ or 3 . Let $u(x, t)$ be the displacement in some direction of the point x at time $t \geq 0$, and the acceleration within V is then

$$\frac{d^2}{dt^2} \int_V u dx = \int_V u_{tt} dx \quad (2.2)$$

while the net force defined on ∂V is given by

$$- \int_{\partial V} \mathbf{F} \cdot \nu dS \quad (2.3)$$

where $\mathbf{F}$ is the force acting on V through ∂V , ν is the outward normal unit vector, then by Newton's second law,

$$\int_V u_{tt} dx = - \int_{\partial V} \mathbf{F} \cdot \nu dS \quad (2.4)$$

where we take the mass of the object to be unit mass, and further this identity obtains for all subregion V so we further have $u_{tt} = -\operatorname{div} \mathbf{F}$. Further for elastic bodies $\mathbf{F}$ is a function of the displacement gradient Du , hence we have the system

$$u_{tt} = \operatorname{div} \mathbf{F}(Du) = 0 \quad (2.5)$$

and for small vibrations we use the approximation formula $\mathbf{F}(Du) = -aDu$ so we have $u_{tt} - a\Delta u = 0$ and wave equation is the case when $a = 1$.

2.2 d'Alembert's Formula

We first focus on the initial value problem for $u(x, t)$ defined on $\mathbb{R} \times (0, \infty)$. Consider the system

$$\begin{cases} u_{tt} - u_{xx} = 0 & \text{in } \mathbb{R} \times (0, \infty) \\ u = g, u_t = h & \text{on } \mathbb{R} \times \{t = 0\} \end{cases} \quad (2.6)$$

Theorem 3. *The general solution to the wave equation $u_{tt} = c^2 u_{xx}$ takes the form*

$$u(x, t) = f(x + ct) + g(x - ct) \quad (2.7)$$

through some functions f, g .

Proof. By direct computation, we have

$$u_x = f'(x + ct) + g'(x - ct), u_{xx} = f''(x + ct) + g''(x - ct) \quad (2.8)$$

and

$$u_t = f'(x + ct) \cdot c + g'(x - ct) \cdot (-c), u_{tt} = c^2 f''(x + ct) + c^2 g''(x - ct) \quad (2.9)$$

and the results follows trivially. ■

Theorem 4. *The d'Alembert's formula is given by*

$$u(x, t) = \frac{1}{2}[g(x + t) + g(x - t)] + \frac{1}{2} \int_{x-t}^{x+t} h(y) dy \quad \text{for } x \in \mathbb{R}, t \geq 0. \quad (2.10)$$

with initial condition $u(x, 0) = g, u_x(x, 0) = h$

Proof. (Evan's) Observe that we can rewrite the PDE as

$$u_{tt} - u_{xx} = \left(\frac{\partial}{\partial t} + \frac{\partial}{\partial x} \right) \left(\frac{\partial}{\partial t} - \frac{\partial}{\partial x} \right) u = 0 \quad (2.11)$$

we define v as

$$v(x, t) := \left(\frac{\partial}{\partial t} - \frac{\partial}{\partial x} \right) u(x, t) \quad (2.12)$$

then we have $v_t(x, t) + v_x(x, t) = 0$. This is a transport equation, where we see the directional derivative along $(1, 1)$ is zero and the function is constant along this constraint. So we have a general solution defined by $v(x, t) = a(x - t)$ for any function a that defined through $x - t$ and $a(x) = v(x, 0)$. Hence we finally have

$$u_t(x, t) - u_x(x, t) = a(x - t) \quad \text{in } \mathbb{R} \times (0, \infty) \quad (2.13)$$

which appears to be another transport equation, the general solution takes the form

$$u(x, t) = \int_0^t a(x + (t - s) - s) ds + b(x + t) \quad (2.14)$$

$$= \frac{1}{2} \int_{x-t}^{x+t} a(y) dy + b(x + t) \quad (2.15)$$

with $b(x) := b(x, 0)$. With initial conditions $b(x) = g(x)$ when $t = 0$, also

$$a(x) = v(x, 0) = u_t(x, 0) - u_x(x, 0) = h(x) - g'(x) \quad (2.16)$$

we finally have

$$u(x, t) = \frac{1}{2} \int_{x-t}^{x+t} (g(y) - g'(y)) dy + g(x + t). \quad (2.17)$$

after simplification we have

$$u(x, t) = \frac{1}{2}[g(x + t) + g(x - t)] + \frac{1}{2} \int_{x-t}^{x+t} h(y) dy, x \in \mathbb{R}, t \geq 0. \quad (2.18)$$

And d'Alembert's formula will give us the solution of the PDE equipped with the boundary value problem. If in general we have $u_{tt} - c^2 u_{xx} = 0$ for some $c \in \mathbb{R}$, then d'Alenbert's formula will yield

$$u(x, t) = \frac{1}{2}[g(x + ct) + g(x - ct)] + \frac{1}{2c} \int_{x-ct}^{x+ct} h(y) dy \quad (2.19)$$

2.3 Energy Conservation

We go back to the 1D wave equation about a vibrating string:

$$u_{tt} - c^2 u_{xx} = 0 \quad (2.20)$$

We first try to derive this formula physically (not via real math). Consider we have an infinity long string with two boundaries fixed, and now we apply a vibration to the string so it will move vertically.



Figure 2: A vibrating string (*Source: Strauss PDE*)

At some moment, the string will look like figure 2. Let x to be a point on the string indicating its position, we denote $u(x, t)$ as the displacement from equilibrium position at time t . If we pick two points $x_0 < x_1$ close enough, and we zoom in into the string and then that part of the string will look like figure 3:

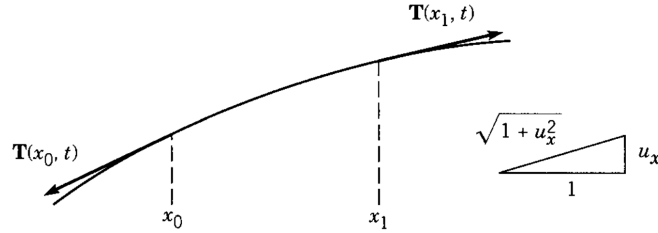


Figure 3: A zoomed vibrating string (*Source: Strauss PDE*)

We assume the string is flexible, elastic and homogeneous with density ρ , so at each point its tension vector is tangential to the string, and we further assume $\mathbf{T}$ does not depend on time t . We use $\mathbf{T}(x_1), \mathbf{T}(x_0)$ to denote the magnitude of the tension vector for the points x_0, x_1 at time t . At point a point x , u_x can be viewed as the tangent line of the string at this point, and the small triangle in figure 2 illustrates this. Since $\mathbf{T}(x)$ is parallel to the tangent line, we can then decompose the tension vector $\mathbf{T}_x$ into two directions $\mathbf{T}_x$ and $\mathbf{T}_y$. Since we assumed the string only moves vertical, so $\mathbf{T}_x = 0$. Then we have

$$\mathbf{T}_y = \mathbf{T}(x) \sin \theta = \mathbf{T}(x) \cdot \frac{u_x}{\sqrt{1 + u_x^2}} \quad (2.21)$$

In segment x_0 to x_1 , the net tension is simply the tension on its boundary since inside tension will cancel out, given by $\mathbf{T}(x) \sin \theta \Big|_{x=x_0}^{x=x_1}$, and at each point its mass times acceleration is just ρu_{tt} , then by integrating over the region, and apply Newton's law, we have

$$\frac{\mathbf{T}(x) u_x}{\sqrt{1 + u_x^2}} \Big|_{x=x_0}^{x=x_1} = \int_{x_0}^{x_1} \rho u_{tt} dx \quad (2.22)$$

when the vibration is small, we immerse the philosophy from the holy engineers and write $\sqrt{1 + u_x^2} = 1$, so we have

$$\mathbf{T}(x_1)u_{x_1} - \mathbf{T}(x_0)u_{x_0} = \int_{x_0}^{x_1} \rho u_{tt} dx \quad (2.23)$$

if we also assume $\mathbf{T}$ does not depend on x , differentiating the term above with respect to x will give us

$$(\mathbf{T}u_x)_x = \rho u_{tt} \quad (2.24)$$

where we let $c = \sqrt{\frac{\mathbf{T}}{\rho}}$ and hence we have

$$\boxed{u_{tt} = c^2 u_{xx}} \quad (2.25)$$

as the wave equation in one dimension.

We now talk about energy conservation in wave equation. Again for simplicity we first consider one-dimensional case.

Definition 5. Let $u_{tt} = c^2 u_{xx}$ be the wave equation of a vibrating string, we define the kinetic energy KE and potential energy PE of the string by the following:

$$KE = \frac{1}{2}\rho \int u_t^2 dx \quad PE = \frac{1}{2}\mathbf{T} \int u_x^2 dx \quad (2.26)$$

Theorem 5. In wave equation the total energy $E = KE + PE$ is conserved.

Proof. We first differentiate the kinetic energy:

$$\frac{d}{dt}KE = \frac{d}{dt} \frac{1}{2}\rho \int u_t^2 dx = \frac{1}{2}\rho \int \frac{d}{dt} u_t^2 dx = \rho \int u_t u_{tt} dx \quad (2.27)$$

then we substitute u_{tt} with $c^2 u_{xx}$ and perform an integration by part:

$$\frac{d}{dt}KE = \rho c^2 \int u_t u_{xx} dx = \mathbf{T} \left(u_t u_x \right)_{x=-\infty}^{x=\infty} - \mathbf{T} \int u_x u_{tx} dx \quad (2.28)$$

where after simplifications we have

$$\frac{d}{dt}KE = -\mathbf{T} \int u_x u_{tx} dx \quad (2.29)$$

finally observe that

$$\frac{d}{dt}PE = \frac{d}{dt} \frac{1}{2}\mathbf{T} \int u_x^2 dx = \mathbf{T} \int u_x u_{xt} dx \quad (2.30)$$

it means we have

$$\frac{d}{dt}(KE + PE) = 0 \quad (2.31)$$

and hence the total energy is conserved. ■

Dirac Delta Function and Distributions

3.1 Basic Definitions and Properties

Definition 6. A distribution F is an operator $F : \varphi \in C_c^\infty(\mathbb{R}) \rightarrow \mathbb{R}$ that is linear and continuous.

For linearity, it means for all $\varphi_1, \varphi_2 \in C_c^\infty(\mathbb{R})$ and $c \in \mathbb{R}$, $F(\varphi_1 + c\varphi_2) = F(\varphi_1) + cF(\varphi_2)$; For continuous it means for all sequence $\{\varphi_n\} \subseteq C_c^\infty(\mathbb{R})$ and $\varphi \in C_c^\infty(\mathbb{R})$, if $\varphi_n \rightarrow \varphi$ in $C_c^\infty(\mathbb{R})$ then $F(\varphi_n) \rightarrow F(\varphi)$. Note that convergence in $C_c^\infty(\mathbb{R})$ means $\varphi_n^{(k)} \rightarrow \varphi^{(k)}$ uniformly for all derivatives $k = 0, 1, \dots$, and uniform convergence means $\forall \varepsilon > 0, \exists N \in \mathbb{N}$ such that $\sup ||\varphi_n^{(k)}(x) - \varphi^{(k)}(x)|| < \varepsilon$ for all x and $n \geq N$.

For notation, we write $\langle F, \varphi \rangle$ for distributions for the action of the distribution F on $\varphi \in C_c^\infty(\mathbb{R})$, and then we can write the linearity as

$$\langle F, \varphi_1 + c\varphi_2 \rangle = \langle F, \varphi_1 \rangle + c \langle F, \varphi_2 \rangle \quad (3.1)$$

Definition 7. Let $f(x)$ be a locally integrable function on $\mathbb{R}$, we define $F_f : C_c^\infty(\mathbb{R}) \rightarrow \mathbb{R}$ by

$$\langle F_f, \varphi \rangle := \int_{\mathbb{R}} f(x)\varphi(x)dx \quad (3.2)$$

for all $\varphi(x) \in C_c^\infty(\mathbb{R})$.

We claim that (3.2) indeed defines a distribution, we can verify its linearity and continuity easily.

Definition 8. Define δ_0 as the distribution $\langle \delta_0, \varphi \rangle = \varphi(0)$ for any $\varphi \in C_c^\infty(\mathbb{R})$.

Example 7. Consider the Heaviside step function $h(x) = \begin{cases} 1 & x \geq 0 \\ 0 & x < 0 \end{cases}$, then

$$\langle F_h, \varphi \rangle = \int_{\mathbb{R}} h(x)\varphi(x)dx = \int_{\mathbb{R}^+} \varphi(x)dx \quad (3.3)$$

is a distribution.

We now consider the derivative of a distribution, for a function $f(x)$ and its derivative $f'(x)$, we can define the following distributions:

$$\langle F_f, \varphi \rangle = \int_{\mathbb{R}} f(x)\varphi(x)dx \quad \langle F_{f'}, \varphi \rangle = \int_{\mathbb{R}} f'(x)\varphi(x)dx \quad (3.4)$$

using integration by parts, we have

$$\int_{\mathbb{R}} f'(x)\varphi(x)dx = f(x)\varphi(x) \Big|_{x=-\infty}^{x=\infty} - \int_{\mathbb{R}} f(x)\varphi'(x)dx = - \int_{\mathbb{R}} f(x)\varphi'(x)dx \quad (3.5)$$

because φ is compactly supported so $\varphi(\infty) = \varphi(-\infty) = 0$. Note that the last term in (3.5) is simply $-\langle F_f, \varphi' \rangle$, so we have the following relation:

$$\langle F_{f'}, \varphi \rangle = -\langle F_f, \varphi' \rangle \quad (3.6)$$

which also motivates the definition of the derivative of a distribution:

Definition 9. Let F be a distribution, then F' is another distribution defined by

$$\langle F', \varphi \rangle := -\langle F, \varphi' \rangle \quad (3.7)$$

for all $\varphi \in C_c^\infty(\mathbb{R})$.

Followed by this definition, we can compute the derivative of Dirac delta function:

$$\langle \delta'_0, \varphi \rangle = -\langle \delta_0, \varphi' \rangle = -\int_{\mathbb{R}} \delta_0 \varphi(x) dx := -\varphi(0). \quad (3.8)$$

Example 8. Consider the function $p(x) = \begin{cases} x & x \geq 0 \\ 0 & x < 0 \end{cases}$, then its distribution is given by

$$\langle F_p, \varphi \rangle = \int_{\mathbb{R}} p(x) \varphi(x) dx = \int_{\mathbb{R}^+} x \varphi(x) dx \quad (3.9)$$

then consider $F_{p'}$, the derivative of such distribution, we have

$$\langle F_{p'}, \varphi \rangle = -\langle F_p, \varphi' \rangle = -\int_{\mathbb{R}^+} x \varphi'(x) dx. \quad (3.10)$$

Then recall the distribution of the Heaviside step function:

$$\langle F_h, \varphi \rangle = \int_{\mathbb{R}^+} \varphi(x) dx := x \varphi(x) \Big|_{x=0}^{x=\infty} - \int_{\mathbb{R}^+} x \varphi'(x) dx = -\int_{\mathbb{R}^+} x \varphi'(x) dx \quad (3.11)$$

using integration by parts and the fact that $\varphi(\infty) = 0$. Then compare (3.10) and (3.9) we see that

$$\langle F_{p'}, \varphi \rangle = \langle F_h, \varphi \rangle. \quad (3.12)$$

Further, what will happen if we consider the derivative of the Heaviside distribution, F'_h ? Let's see:

$$\langle F_h, \varphi \rangle = \langle F'_H, \varphi \rangle = -\langle F_H, \varphi' \rangle = -\int_{\mathbb{R}^+} 1 \cdot \varphi'(x) dx = \varphi(0) \quad (3.13)$$

since again $\varphi(\infty) = 0$. Then by definition, we have

$$\langle F_h, \varphi \rangle = \langle \delta_0, \varphi \rangle. \quad (3.14)$$

That is, the derivative of the Heaviside function in the sense of distributions is the Dirac delta function.

Example 9. Let $f(x) = |x|$, which is, non-differentiable at $x = 0$.

We will try differentiations in the sense of distributions:

$$\langle F'_f, \varphi \rangle = -\langle F_f, \varphi' \rangle = \int_{\mathbb{R}} |x| \varphi'(x) dx = -\left(\int_{-\infty}^0 -x \varphi'(x) dx + \int_0^{\infty} x \varphi'(x) dx \right) \quad (3.15)$$

which is,

$$\langle F'_f, \varphi \rangle = x \varphi(x) \Big|_{x=-\infty}^0 - \int_{-\infty}^0 \varphi(x) dx - x \varphi(x) \Big|_{x=0}^{x=\infty} + \int_0^{\infty} \varphi(x) dx \quad (3.16)$$

and after simplifications we have

$$\langle F'_f, \varphi \rangle = \int_{\mathbb{R}} g(x) \varphi(x) dx = \langle F_g, \varphi \rangle \quad (3.17)$$

where $g(x) = \begin{cases} -1 & x < 0 \\ 1 & x \geq 0 \end{cases}$.

Example 10. Let $f(x) = \begin{cases} x^2 & x \geq 0 \\ 2x + 3 & x < 0 \end{cases}$, then what is f' in the sense of distributions?

Let us compute directly from the definition:

$$\langle F'_f, \varphi \rangle = -\langle F_f, \varphi' \rangle \quad (3.18)$$

$$= -\int_{-\infty}^0 (2x + 3) \varphi'(x) dx - \int_0^{\infty} x^2 \varphi'(x) dx \quad (3.19)$$

$$= -\left((2x + 3) \varphi(x) \Big|_{x=-\infty}^0 - \int_{-\infty}^0 2\varphi(x) dx + x^2 \varphi(x) \Big|_{x=0}^{x=\infty} - \int_0^{\infty} 2x \varphi(x) dx \right) \quad (3.20)$$

$$= -\left(3\varphi(0) - 2 \int_{-\infty}^0 \varphi(x) dx - \int_0^{\infty} 2x \varphi(x) dx \right) \quad (3.21)$$

$$= \int_{-\infty}^{\infty} g(x) \varphi(x) dx - 3\varphi(0) \quad (3.22)$$

$$:= \langle g, \varphi \rangle - 3\varphi(0) \quad (3.23)$$

$$= \langle g - 3\delta_0, \varphi \rangle. \quad (3.24)$$

where $g(x) = \begin{cases} 2 & x < 0 \\ 2x & x \geq 0 \end{cases}$. We say that the derivative of f in the sense of distributions is the distribution $g - 3\delta_0$, or $F_g - 3\delta_0$. The $3\delta_0$ is a result of the jump discontinuity in f at $x = 0$.

Let us consider a more general case of discontinuity. Define

$$f(x) = \begin{cases} h(x) & x < 0 \\ g(x) & x > 0 \end{cases} \quad (3.25)$$

where $g(0) - h(0) := c$ has a jump hence $f(x)$ is discontinuous at 0. We now consider the derivative of f in sense of distributions. By direct computation, we have

$$\langle F'_f, \varphi \rangle = -\langle F_f, \varphi' \rangle \quad (3.26)$$

$$= -\left(\int_{-\infty}^0 h(x) \varphi'(x) dx + \int_0^{\infty} g(x) \varphi'(x) dx \right) \quad (3.27)$$

$$= -\left(h(x) \varphi(x) \Big|_{-\infty}^0 - \int_{-\infty}^0 h'(x) \varphi(x) dx - g(x) \varphi(x) \Big|_0^{\infty} - \int_0^{\infty} g'(x) \varphi(x) dx \right) \quad (3.28)$$

where further simplification shows that

$$\langle F'_f, \varphi \rangle = (g(0) - h(0))\varphi(0) + \int_{-\infty}^0 h'(x)\varphi(x)dx + \int_0^{\infty} g'(x)\varphi(x)dx \quad (3.29)$$

$$:= \langle c\delta_0, \varphi \rangle + \langle F'_f, \varphi \rangle \quad (3.30)$$

where $f'(x) = \begin{cases} h'(x) & x < 0 \\ g'(x) & x > 0 \end{cases}$, and $c\delta_0$ just represents that “jump discontinuity”.

3.2 Convergence of Functions in Distributions

Definition 10. A sequence F_n of distributions converges to a distribution F if

$$\lim_{n \rightarrow \infty} \langle F_n, \varphi \rangle = \langle F, \varphi \rangle \quad \text{for all } \varphi \in C_C^\infty(\mathbb{R}). \quad (3.31)$$

We may consider a sequence of locally integrable functions $\{f_n\}$, and we can define its convergence in the sense of distribution:

Definition 11. A sequence of locally integrable functions $\{f_n\}$ converges to f in the sense of distributions if

$$\lim_{n \rightarrow \infty} \langle F_{f_n}, \varphi \rangle = \langle F_f, \varphi \rangle \quad \text{for all } \varphi \in C_C^\infty(\mathbb{R}), \quad (3.32)$$

which is,

$$\lim_{n \rightarrow \infty} \int_{\mathbb{R}} f_n(x)\varphi(x)dx = \int_{\mathbb{R}} f(x)\varphi(x)dx \quad \text{for all } \varphi \in C_C^\infty(\mathbb{R}). \quad (3.33)$$

Proposition 1. Let $\{F_n\}, F$ be distributions, if $F_n \rightarrow F$, then $F_n^{(\alpha)} \rightarrow F^{(\alpha)}$ for all higher order derivatives.

The proof is easy since $\varphi(x) \in C_C^\infty(\mathbb{R})$.

Proposition 2. Let $\{f_n(x)\} \subseteq L_{loc}^1$ be a sequence of locally integrable functions and $f_n \rightarrow f$ pointwise where $f \in L_{loc}^1$. Further suppose $\exists g \in L_{loc}^1$ such that $|f_n(x)| \leq g(x)$ for all n and $x \in \mathbb{R}$, then $f_n \rightarrow f$ in the sense of distributions.

The proof follows from monotone convergence theorem and dominated convergence theorem. An application to the proposition above is to consider the sequence

$$f_n(x) = \frac{1}{n}e^{-x^2/4n} \quad (3.34)$$

and it can be shown that $f_n(x) \rightarrow 0$ pointwise, so $f_n(x) \rightarrow 0$ in the sense of distributions.

Definition 12. A sequence $\{f_n\} \subseteq L_{loc}^1$ converges in the sense of distributions to δ_0 (the Dirac delta function), if

$$\int_{\mathbb{R}} f_n(x)\varphi(x)dx \xrightarrow{n \rightarrow \infty} \varphi(0) \quad \text{for all } \varphi \in C_C^\infty(\mathbb{R}). \quad (3.35)$$

Proposition 3. *If a sequence $\{f_n\} \subseteq L^1_{loc}$ converges to δ_0 in distribution, then $f_n(x)$ satisfies the following properties:*

(1) **Nonnegativity:** $f_n(x) \geq 0$ for all $x \in \mathbb{R}, n \in \mathbb{N}$;

(2) **Unit mass:** We have $\int_{\mathbb{R}} f_n(x) dx = 1$ for all $n \in \mathbb{N}$.

(3) **Concentration at 0:** For any $\varepsilon > 0$, $f_n(x) \rightarrow 0$ uniformly on $|x| \geq \varepsilon$.

Theorem 6. *Let $f(x)$ be any non-negative integrable function on $\mathbb{R}$ and $\int_{\mathbb{R}} f(x) dx = 1$. Define $f_n := nf(nx)$, then $f_n \rightarrow \delta_0$ in distribution.*

Proof. We need to show $\langle f_n, \varphi \rangle \rightarrow \langle \delta_0, \varphi \rangle := \varphi(0)$, i.e. $\int_{\mathbb{R}} f_n(x) \varphi(x) dx \rightarrow \varphi(0)$ in distribution. Using $\varepsilon - \delta$ definition, $\forall \varepsilon > 0$, since f_n is non-negative, we have

$$\left| \int_{\mathbb{R}} f_n(x) \varphi(x) dx - \varphi(0) \right| = \left| \int_{\mathbb{R}} f_n(x) \cdot (\varphi(x) - \varphi(0)) dx \right| \quad (3.36)$$

$$\leq \int_{\mathbb{R}} f_n(x) \cdot \left| \varphi(x) - \varphi(0) \right| dx \quad (3.37)$$

$$= \int_{\mathbb{R}} f(y) \cdot \left| \varphi\left(\frac{y}{n}\right) - \varphi(0) \right| dy \quad (3.38)$$

By continuity of φ , $\forall \varepsilon > 0$, one can choose δ , such that for $N \in \mathbb{N}$ sufficiently large, $\left| \frac{y}{n} - 0 \right| < \delta$ for all $n \geq N$ would imply $\left| \varphi\left(\frac{y}{n}\right) - \varphi(0) \right| < \varepsilon$, hence

$$\int_{\mathbb{R}} f(y) \cdot \left| \varphi\left(\frac{y}{n}\right) - \varphi(0) \right| dy \leq \varepsilon \int_{\mathbb{R}} f(y) dy \equiv \varepsilon. \quad (3.39)$$

for all $n \geq N$, and hence

$$\left| \int_{\mathbb{R}} f_n(x) \varphi(x) dx - \varphi(0) \right| < \varepsilon \quad (3.40)$$

for all $n \geq N$ which means $f_n(x) \rightarrow \delta_0$ in distribution. ■

Theorem 7. $\sin(nx) \rightarrow 0$ in distribution.

Proof. Note that $\varphi \in C^\infty_c(\mathbb{R})$, so $\exists M \in \mathbb{R}^+$ such that $\varphi(x) = 0$ for all $|x| \geq M$, hence

$$\int_{\mathbb{R}} \sin(nx) \varphi(x) dx = \int_{[-M, M]} \sin(nx) \varphi(x) dx. \quad (3.41)$$

Then we have

$$\left| \int_{[-M,M]} \sin(nx) \varphi(x) dx - 0 \right| = \left| \int_{[-M,M]} \left(-\frac{d}{dx} \cdot \frac{\cos(nx)}{n} \right) \varphi(x) dx \right| \quad (3.42)$$

$$= \left| -\frac{\cos(nx)}{n} \cdot \varphi(x) \Big|_{x=-M}^{x=M} + \int_{[-M,M]} \frac{\cos(nx)}{n} \cdot \varphi'(x) dx \right| \quad (3.43)$$

$$\leq \frac{1}{n} \int_{[-M,M]} \left| \cos(nx) \cdot \varphi'(x) \right| dx \quad (3.44)$$

$$\leq \frac{1}{n} \int_{[-M,M]} \|\varphi'(x)\|_{L^\infty} dx \quad (3.45)$$

$$= \frac{2M \|\varphi'(x)\|_{L^\infty}}{n}. \quad (3.46)$$

Simply let $\varepsilon := \frac{2M \|\varphi'(x)\|_{L^\infty}}{n}$, choose N such that $N > \frac{2M \|\varphi'(x)\|_{L^\infty}}{\varepsilon}$ so for all $n \geq N$, we have equation (3.36) $< \varepsilon$. $\blacksquare$

Example 11. Show that the sequence of function defined by $f_n(x) = \frac{1}{\pi} \cdot \frac{n}{n^2 x^2 + 1}$ converges to δ_0 in distribution.

Proof. By definition we wish to show that $\forall \varepsilon > 0, \exists N \in \mathbb{N}$ such that

$$\left| \int_{\mathbb{R}} f_n(x) \varphi(x) dx - \varphi(0) \right| < \varepsilon \quad (3.47)$$

for all $n \geq N$. By a similar argument in the proof of **theorem 6** we obtain

$$\left| \int_{\mathbb{R}} f_n(x) \varphi(x) dx - \varphi(0) \right| \leq \int_{\mathbb{R}} f_n(x) \cdot \left| \varphi(x) - \varphi(0) \right| dx. \quad (3.48)$$

Since $\varphi \in C_c^\infty(\mathbb{R})$, so for the ε given, $\exists \delta$ such that $|x| < \delta$ implies $|\varphi(x) - \varphi(0)| < \varepsilon$. We then rewrite the right hand side of (3.42) as

$$\int_{\mathbb{R}} f_n(x) \cdot \left| \varphi(x) - \varphi(0) \right| dx = \underbrace{\int_{[-\delta, \delta]} f_n(x) \cdot \left| \varphi(x) - \varphi(0) \right| dx}_I + \underbrace{\int_{\mathbb{R}/[-\delta, \delta]} f_n(x) \cdot \left| \varphi(x) - \varphi(0) \right| dx}_J \quad (3.49)$$

where we will investigate I, J separately. For I term, use the fact that $x \in [-\delta, \delta]$, we have

$$I < \varepsilon \int_{[-\delta, \delta]} f_n(x) dx \leq \varepsilon \quad (3.50)$$

since we have $\int_{\mathbb{R}} f_n(x) dx = 1$. For J , we have

$$J \leq \int_{\mathbb{R}/[-\delta, \delta]} f_n(x) \cdot \left(|\varphi(x)| + |\varphi(0)| \right) dx \quad (3.51)$$

$$\leq \int_{\mathbb{R}/[-\delta, \delta]} f_n(x) \cdot 2 \|\varphi(x)\|_{L^\infty} dx \quad (3.52)$$

$$:= \frac{4 \|\varphi(x)\|_{L^\infty}}{\pi} \int_{\delta}^{\infty} \frac{n}{n^2 x^2 + 1} dx \quad (3.53)$$

$$= \frac{4 \|\varphi(x)\|_{L^\infty}}{\pi} \cdot \left(\frac{\pi}{2} - \arctan(n\delta) \right) \quad (3.54)$$

by choosing N sufficiently large, we have $\frac{\pi}{2} - \arctan(n\delta) < \varepsilon$ for all $n \geq N$. So all combined, we have

$$\left| \int_{\mathbb{R}} f_n(x) \varphi(x) - \varphi(0) \right| < 2\varepsilon \quad (3.55)$$

for any $\varepsilon > 0$ and sufficiently large $n > N$. Hence $f_n(x) \rightarrow \delta_0$ in distribution. ■

Laplace Equations

4.1 Fundamental Solutions

Definition 13. Let $u(x_1, \dots, x_n) : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ be a unknown function, the Laplacian of the function is defined by

$$\Delta u = \sum_{i=1}^n \frac{\partial^2 u}{\partial x_i^2}. \quad (4.1)$$

Laplace equation is of the form

$$\Delta u = 0 \quad (4.2)$$

for all $x \in U$, any function satisfying the above is also called harmonic.

Lemma 1. Laplace's equation is invariant under rotation, i.e if P is an orthogonal matrix with entries p_{ij} , then $\Delta u(Px) = 0$ for all $x \in U$.

Proof. By Chain rule, we have

$$D_{x_i}(u(Px)) = \sum_{k=1}^n D_{x_k}(u(Px)p_{ik}) \quad (4.3)$$

and similarly

$$D_{x_i x_j}(u(Px)) = \sum_{l=1}^n \sum_{k=1}^n D_{x_k x_l}(u(Px)p_{ik}p_{jl}). \quad (4.4)$$

Since P is orthogonal, thus $p_{ik}p_{il} = 1$ iff $k = l$ and hence

$$\Delta u(Px) = \sum_{i=1}^n \sum_{l=1}^n \sum_{k=1}^n D_{x_k x_l}(u(Px)p_{ik}p_{il}) = \Delta u = 0. \quad (4.5)$$

■

Given this, we seek for radial solutions which takes the form $u(x) = v(r)$ where $r = |x|$. Denote $r = (x_1^2 + \dots + x_n^2)^{1/2}$, let $x \neq 0$, then we see that

$$\frac{\partial r}{\partial x_i} = (x_1^2 + \dots + x_n^2)^{-1/2} x_i = \frac{x_i}{r} \quad (4.6)$$

and by Chain rule we have

$$u_{x_i} = v'(r) \frac{x_i}{r} \quad (4.7)$$

$$u_{x_i x_i} = v''(r) \frac{x_i^2}{r^2} + v'(r) \frac{r - x_i \frac{x_i}{r}}{r^2} = v''(r) \frac{x_i^2}{r^2} + v'(r) \left(\frac{1}{r} - \frac{x_i^2}{r^3} \right) \quad (4.8)$$

using the fact that $x_1^2 + \dots + x_n^2 = r^2$, we would have

$$\Delta u = \sum_{i=1}^n \left(v''(r) \frac{x_i^2}{r^2} + v'(r) \left(\frac{1}{r} - \frac{x_i^2}{r^3} \right) \right) = v''(r) + \frac{n-1}{r} v'(r). \quad (4.9)$$

by letting $\Delta u = 0$, we obtain the following ODE:

$$v''(r) + \frac{n-1}{r}v'(r) = 0. \quad (4.10)$$

Now if $v'(r) \neq 0$, we see that

$$\log(v'(r))' = \frac{v''(r)}{v'(r)} = \frac{1-n}{r} \quad (4.11)$$

and we integrate with respect to r , we have

$$\log(v'(r)) = (1-n)\log r + C \quad (4.12)$$

which simplifies to $v'(r) = C_0/r^{n-1}$ for some constant C_0 , hence the general solution $v(r)$ takes the form

$$v(r) = \begin{cases} A \log r + B & n = 2 \\ \frac{C}{r^{n-2}} + D & n \geq 3 \end{cases} \quad (4.13)$$

where A, B, C, D are all constants. This motivates the fundamental solution to Laplace's equation:

Definition 14. *The function defined by*

$$\Phi(x) = \begin{cases} -\frac{1}{2\pi} \log |x| & n = 2 \\ \frac{1}{n(n-2)\alpha(n)} \cdot \frac{1}{|x|^{n-2}} & n \geq 3 \end{cases} \quad (4.14)$$

is the fundamental solution to Laplace's equation $\Delta u = 0$ for all $x \in \mathbb{R}^n/\{0\}$ where $\alpha(n)$ denotes the volume of the unit ball in $\mathbb{R}^n$.

Laplace's equation is a special class of Poisson's equations, we define Poisson's equation as

Definition 15. *For a given $f : \mathbb{R}^n \rightarrow \mathbb{R}$, and the unknown $u(x_1, \dots, x_n) : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$, the Poisson's equation takes the form*

$$-\Delta u = f. \quad (4.15)$$

By construction, we know $\Phi(x)$ is harmonic since it solves $\Delta u = 0$. We shift the origin to a new point y , and under invariant we still have $\Delta u = 0$ and now $\Phi(x - y)$ is harmonic in x . Consider a given function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and that $\Phi(x - y)f(y)$ is also harmonic when $x \neq y$, by taking all possible y , it suggests the following convolution integral:

$$u(x) = \int_{\mathbb{R}^n} \Phi(x - y)f(y)dy. \quad (4.16)$$

We may want to compute Δu in this case but we see that it is hard to deal with at when $x = y$, the next theorem will formally prove $u(x)$ is indeed the fundamental solution to the Poisson's equation:

Theorem 8. *Define $u(x) = \int_{\mathbb{R}^n} \Phi(x - y)f(y)dy$ and assume that $f \in C_c^2(\mathbb{R}^n)$, then:*

- (i) $u \in C^2(\mathbb{R}^n)$;
- (ii) $-\Delta u = f$ in $\mathbb{R}^n$.

We will prove both results separately:

For (i):

Proof. Use the property of a convolution, we have

$$u(x) = \int_{\mathbb{R}^n} \Phi(y) f(x-y) dy \quad (4.17)$$

then denote e_i as the unit vector in i th position of $x = (x_1, \dots, x_n)$, and abusing the notion of interchanging limits and integrals (we actually could do that), we have

$$\begin{aligned} \frac{\partial u}{\partial x_i}(x) &= \lim_{h \rightarrow 0} \frac{u(x + he_i) - u(x)}{h} = \int_{\mathbb{R}^n} \Phi(y) \cdot \lim_{h \rightarrow 0} \left[\frac{f(x + he_i - y) - f(x - y)}{h} \right] dy \\ &= \int_{\mathbb{R}^n} \Phi(y) \cdot \frac{\partial f}{\partial x_i}(x - y) dy. \end{aligned}$$

Similarly one can show

$$\frac{\partial u}{\partial x_i \partial x_j}(x) = \int_{\mathbb{R}^n} \Phi(y) \cdot \frac{\partial^2 f}{\partial x_i \partial x_j}(x - y) dy. \quad (4.18)$$

Since $f \in C_C^2(\mathbb{R}^n)$, the second partial derivative is continuous w.r.t x , hence $u \in C^2(\mathbb{R}^n)$. ■

For (ii):

Proof. We split the integral into two parts to avoid the blow-up of Φ near 0, $\forall \varepsilon > 0$ fixed, denote $B(0, \varepsilon)$ as the ball centered at 0 with radius ε , write

$$u(x) = \int_{B(0, \varepsilon)} \Phi(y) f(x-y) dy + \int_{\mathbb{R}^n / B(0, \varepsilon)} \Phi(y) f(x-y) dy \quad (4.19)$$

hence

$$\Delta u(x) = \int_{B(0, \varepsilon)} \Phi(y) \Delta_x f(x-y) dy + \int_{\mathbb{R}^n / B(0, \varepsilon)} \Phi(y) \Delta_x f(x-y) dy. \quad (4.20)$$

Denote the first term as I_ε , second term as J_ε . Our first goal is to bound I_ε , proceed as follows:

When $n = 2$, we have

$$|I_\varepsilon| = \left| - \int_{B(0, \varepsilon)} \frac{1}{2\pi} \log |y| \cdot \Delta_x f(x-y) dy \right| \leq C \|Df\|_{L^\infty(\mathbb{R}^n)} \cdot \left| \int_{B(0, \varepsilon)} \log |y| dy \right|$$

for some constant C , and we further have

$$\int_{B(0, \varepsilon)} \log |y| dy = \int_0^\varepsilon \int_0^{2\pi} \log |r| r d\theta dr = \varepsilon^2 \log |\varepsilon| \quad (4.21)$$

so we have $|I_\varepsilon| < C\varepsilon^2|\log \varepsilon|$ for some constant C . When $n \geq 3$, similarly we can show that $|I_\varepsilon| < C\varepsilon$, hence the term I_ε is bounded in terms of arbitrary ε . Now for J_ε , first note that $\Delta_x f(x-y) = \Delta_y f(x-y)$, we have

$$J_\varepsilon = \int_{\mathbb{R}^n/B(0,\varepsilon)} \Phi(y) \Delta_y f(x-y) dy. \quad (4.22)$$

To compute this integral, we need some technical lemmas:

Lemma 2. (*Gauss-Green Theorem*) Let $u(x_1, \dots, x_n) \in C^1(\overline{U})$, let $\nu = (\nu_1, \dots, \nu_n)$ be the outward pointing unit normal vector field of U defined on ∂U then

$$\int_U u_{x_i} dx = \int_{\partial U} u \nu_i dS. \quad (4.23)$$

We can use this lemma to further derive an integration by parts formula:

Lemma 3. (*Integration by Parts*) Let $u, v \in C^1(\overline{U})$, then

$$\int_U u_{x_i} v dx = - \int_U u v_{x_i} dx + \int_{\partial U} u v \nu_i dS. \quad (4.24)$$

Then we may rewrite J_ε as

$$J_\varepsilon = - \int_{\mathbb{R}^n/B(0,\varepsilon)} D\Phi(y) \cdot D_y f(x-y) dy + \int_{\partial B(0,\varepsilon)} \Phi(y) \nu Df(x-y) dS(y) \quad (4.25)$$

where ν is the inward pointing unit normal vector field. We now denote the first term by K_ε and the second term by L_ε , then using the same technique as we did for I_ε , we can check that

$$|L_\varepsilon| \leq C \|Df\|_{L^\infty(\mathbb{R}^n)} \int_{\partial B(0,\varepsilon)} |\Phi(y)| dS(y) \leq \begin{cases} C\varepsilon |\log \varepsilon| & n = 2 \\ C\varepsilon & n \geq 3 \end{cases} \quad (4.26)$$

hence L_ε is also bounded in terms of ε . We again perform integration by parts in K_ε , and we have

$$\begin{aligned} K_\varepsilon &= \int_{\mathbb{R}^n/B(0,\varepsilon)} \Delta \Phi(y) f(x-y) dy - \int_{\partial B(0,\varepsilon)} \nu D\Phi(y) f(x-y) dS(y) \\ &= - \int_{\partial B(0,\varepsilon)} \nu D\Phi(y) f(x-y) dS(y). \end{aligned}$$

Since $D\Phi(y) = \frac{-1}{n\alpha(n)} \frac{y}{|y|^n}$ when $y \neq 0$ and $\nu = -y/|y| = -y/\varepsilon$ on $\partial B(0,\varepsilon)$, so we will have

$$\nu D\Phi(y) = \frac{1}{n\alpha(n)\varepsilon^{n-1}} \quad (4.27)$$

also $n\alpha(n)\varepsilon^{n-1}$ is the surface area of the ball $\partial B(0,\varepsilon)$, so

$$K_\varepsilon = - \frac{1}{n\alpha(n)\varepsilon^{n-1}} \int_{\partial B(0,\varepsilon)} f(x-y) dS(y) = - \oint_{\partial B(0,\varepsilon)} f(y) dS(y) \rightarrow -f(x) \quad (4.28)$$

as $\varepsilon \rightarrow 0$. Since we have shown that $I_\varepsilon, L_\varepsilon$ are all bounded by terms of ε so by setting $\varepsilon \rightarrow 0$ they will vanish as well. Hence we have

$$-\Delta u(x) = f(x). \quad (4.29)$$

■

4.2 Properties of Harmonic Functions

4.2.1 Mean Value Property

For a harmonic function u , its value $u(x)$ is equal to the average u over both the boundary $\partial B(x, r)$ and the ball $B(x, r)$ as long as $B(x, r) \in U$.

Theorem 9. *Let $U \subset \mathbb{R}^n$ open, if $u(x_1, \dots, x_n) \in C^2(U)$ is harmonic then for each ball $B(x, r) \in U$,*

$$u(x) = \oint_{\partial B(x, r)} u(y) dS(y) = \int_{B(x, r)} u(y) dy \quad (4.30)$$

where

$$\oint_{\partial B(x, r)} u(y) dS(y) = \frac{1}{n\alpha(n)r^{n-1}} \int_{\partial B(x, r)} u(y) dS(y), \quad (4.31)$$

$$\int_{B(x, r)} u(y) dy = \frac{1}{\alpha(n)r^n} \int_{B(x, r)} u(y) dy \quad (4.32)$$

$\alpha(n)$ is the volume of the unit ball in $\mathbb{R}^n$ and $n\alpha(n)$ is the surface area of the unit ball in $\mathbb{R}^n$.

Proof. Let $u(x_1, \dots, x_n) \in C^2(U)$ be harmonic, and define

$$\varphi(r) = \begin{cases} \oint_{\partial B(x, r)} u(y) dS(y) & r > 0 \\ u(x) & r = 0 \end{cases}. \quad (4.33)$$

If u is a smooth function (we later will show all harmonic functions are smooth), then $\lim_{r \rightarrow 0} \varphi(r) = u(x)$ and hence φ is continuous. Note that by change of variables, we have

$$\varphi(r) = \int_{\partial B(0, 1)} u(x + rz) dS(z) \quad (4.34)$$

and we take the derivative of $\varphi(r)$:

$$\begin{aligned} \varphi'(r) &= \int_{\partial B(0, 1)} \nabla u(x + rz) \cdot z dS(z) \\ &= \int_{\partial B(x, r)} \nabla u(y) \cdot \frac{y - x}{r} dS(y) \\ &= \int_{\partial B(x, r)} \frac{\partial u}{\partial \nu}(y) dS(y) \\ &= \frac{1}{n\alpha(n)r^{n-1}} \int_{\partial B(x, r)} \frac{\partial u}{\partial \nu}(y) dS(y) \\ &= \frac{1}{n\alpha(n)r^{n-1}} \int_{B(x, r)} \nabla(\nabla u) dy \\ &= \frac{1}{n\alpha(n)r^{n-1}} \int_{B(x, r)} \Delta u(y) dy \equiv 0. \end{aligned}$$

which means φ is a constant function, and hence we have

$$u(x) = \oint_{\partial B(x,r)} u(y) dS(y). \quad (4.35)$$

Furthermore we have

$$\begin{aligned} \int_{B(x,r)} u(y) dy &= \int_0^r \left(\int_{\partial B(x,s)} u dS \right) ds \\ &= u(x) \int_0^r n\alpha(n)s^{n-1} ds \\ &= \alpha(n)r^n u(x). \end{aligned}$$

Another result is the converse to mean-value property:

Theorem 10. *If $u \in C^2(U)$ satisfies*

$$u(x) = \oint_{\partial B(x,r)} u dS \quad (4.36)$$

for each ball $B(x,r) \in U$, then u is harmonic.

Proof. Suppose $\Delta u \neq 0$, then $\exists B(x,r) \in U$ such that $\Delta u > 0$ say, but then

$$\varphi'(r) = \frac{1}{n\alpha(n)r^{n-1}} \int_{\partial B(x,r)} \Delta u dS = \frac{r}{n} \oint_{B(x,r)} \Delta u(y) dy > 0 \quad (4.37)$$

which contradicts the fact that φ is a constant function. ■

4.2.2 Maximum Principle

A nice property of harmonic functions is that if u is harmonic on a bounded domain Ω , then u attains its maximum value on the boundary of Ω .

Theorem 11. *(Strong Maximum Principle) Suppose $u \in C^2(U) \cap C(\bar{U})$ is harmonic within U , then*

(i) $\max_{\bar{U}} u = \max_{\partial U} u$;

(ii) Furthermore, if U is connected and there exists a point $x_0 \in U$ such that $u(x_0) = \max_{\bar{U}} u$, then u is constant within U .

Proof. We will prove the second statement since the first followed from the second. Suppose $\exists x_0 \in U$ such that $u(x_0) = \max_{\bar{U}} u(x) \equiv M$. Then $\forall r \in (0, \text{dist}(x_0, \partial U))$, the mean-value property states that

$$M = u(x_0) = \oint_{B(x_0,r)} u(y) dy \quad (4.38)$$

by taking the "average" around the maximum x_0 , we would expect

$$\oint_{B(x_0,r)} u(y) dy \leq M \quad (4.39)$$

and such will hold only if $u \equiv M$ within $B(x_0, r)$, and $u(y) = M$. ■

An important application of maximum principle is establishing the uniqueness of solutions to certain boundary value problems for Poisson's equation.

Theorem 12. *Let $g \in C(\partial U)$, $f \in C(U)$, then there exists at most one solution $u \in C^2(U) \cap C(\bar{U})$ of the boundary value problem*

$$\begin{cases} -\Delta u = f & \text{in } U \\ u = g & \text{on } \partial U \end{cases} \quad (4.40)$$

Proof. Assume u_1, u_2 both solves the boundary value problem, then let $w = u_1 - u_2$, then we have

$$\begin{cases} -\Delta w = 0 & \text{in } U \\ w = 0 & \text{in } \partial U \end{cases} \quad (4.41)$$

then by maximum principle, $\max_{\partial U} w = \max_{\partial \bar{U}} w = 0$, i.e $u_1 = u_2$. ■

4.2.3 Smoothness

A nice thing about harmonic functions is that they are smooth!

Theorem 13. *If $u \in C(U)$ satisfies the mean-value property for each ball $B(x, r) \in U$, then $u \in C^\infty(U)$.*

The proof of this theorem requires some knowledge on mollifiers, so before the proof we shall spend some time on mollifiers.

Definition 16. *Define a smooth function $\eta \in C^\infty(\mathbb{R}^n)$ by*

$$\eta(x) = \begin{cases} C \exp\left(\frac{1}{|x|^2-1}\right) & |x| < 1 \\ 0 & |x| \geq 1 \end{cases} \quad (4.42)$$

for some constant $C > 0$ so that $\int_{\mathbb{R}^n} \eta dx = 1$. Then for each $\varepsilon > 0$, set

$$\eta_\varepsilon(x) = \frac{1}{\varepsilon^n} \eta\left(\frac{x}{\varepsilon}\right). \quad (4.43)$$

We call η the standard mollifier, the functions η_ε are C^∞ as well.

Definition 17. *Denote $U_\varepsilon = \{x \in U \mid \text{dist}(x, \partial U) > \varepsilon\}$, if a function $f : U \rightarrow \mathbb{R}$ is locally integrable, its mollification is defined by*

$$f^\varepsilon = \int_U \eta_\varepsilon(x - y) f(y) dy \quad (4.44)$$

for $x \in U_\varepsilon$.

Now we prove the theorem:

Proof. Let η be the standard mollifier, set $u^\varepsilon = \eta_\varepsilon * u$ in U_ε , we first show $u^\varepsilon \in C^\infty$, which is left as an exercise to the reader. Then to show u is smooth, we will show in fact $u \equiv u^\varepsilon$ on U_ε for each $\varepsilon > 0$. Let $x \in U_\varepsilon$, then we know the support of $\eta_\varepsilon(x - y)$ as a function of y is given by $\{y : |x - y| < \varepsilon\}$, and thus we have

$$\begin{aligned}
u^\varepsilon(x) &= \frac{1}{\varepsilon^n} \int_{B(x,\varepsilon)} \eta\left(\frac{|x-y|}{\varepsilon}\right) u(y) dy \\
&= \frac{1}{\varepsilon^n} \int_0^\varepsilon \eta\left(\frac{r}{\varepsilon}\right) \left(\int_{\partial B(x,r)} u dS \right) dr \\
&= \frac{1}{\varepsilon^n} u(x) \int_0^\varepsilon \eta\left(\frac{r}{\varepsilon}\right) n\alpha(n)r^{n-1} dr \\
&= u(x) \int_{B(0,\varepsilon)} \eta_\varepsilon dy \\
&= u(x).
\end{aligned}$$

so $u \in C^\infty(U_\varepsilon)$ for all $\varepsilon > 0$. ■

4.2.4 Liouville's Theorem

Theorem 14. *If $u : \mathbb{R}^n \rightarrow \mathbb{R}$ is harmonic and bounded (either from above or below), then u is constant.*

To prove Liouville's theorem, we first introduce a technical lemma for estimating the derivatives of harmonic functions:

Lemma 4. *Assume u is harmonic in U , then for multi-index $\alpha = 1$,*

$$|D^\alpha u(x_0)| \leq \frac{n}{\alpha(n)} \cdot \left(\frac{2}{r}\right)^{n+1} \cdot \|u\|_{L^1(B(x_0,r))}. \quad (4.45)$$

for each ball $B(x_0, r) \in U$.

Proof. Let $x_0 \in U$ and $B(x, r) \subseteq U$. Since u_{x_i} is also harmonic, then use mean-value property we have

$$\begin{aligned}
|u_{x_i}(x_0)| &= \left| \frac{2^n}{\alpha(n)r^n} \int_{B(x_0, \frac{r}{2})} u_{x_i}(y) dy \right| \\
&= \left| \frac{2^n}{\alpha(n)r^n} \int_{\partial B(x_0, \frac{r}{2})} u(y) \nu^i dS(y) \right| \\
&= \left| \frac{2n}{r} \int_{\partial B(x_0, \frac{r}{2})} u(y) \nu^i dS(y) \right| \\
&\leq \frac{2n}{r} \|u\|_{L^\infty(\partial B(x_0, r/2))}.
\end{aligned}$$

Now for $u(x_0)$, note that

$$|u(x_0)| = \left| \frac{2^n}{\alpha(n)r^n} \int_{B(x_0, \frac{r}{2})} u(y) dy \right| \leq \frac{2^n}{\alpha(n)r^n} \|u\|_{L^1(B(x_0, r))}. \quad (4.46)$$

Hence we have

$$|D^\alpha u(x_0)| \leq \frac{2n}{r} \|u\|_{L^\infty(\partial B(x_0, r/2))} \leq \left(\frac{2}{r}\right)^{n+1} \cdot \frac{n}{\alpha(n)} \|u\|_{L^1(B(x_0, r))}. \quad (4.47)$$
■

We now prove Liouville's theorem:

Proof. Fix $x_0 \in \mathbb{R}^n$ and $r > 0$, we have

$$|Du(x_0)| \leq \frac{n}{\alpha(n)} \left(\frac{2}{r}\right)^{n+1} \cdot \|u\|_{L^1(B(x_0, r))} \leq \frac{n}{\alpha(n)} \left(\frac{2}{r}\right)^{n+1} \cdot \|u\|_{L^\infty(\mathbb{R}^n)} \xrightarrow{r \rightarrow \infty} 0. \quad (4.48)$$

which means u is a constant. ■

Theorem 15. *Let $f \in C_C^2(\mathbb{R}^n)$ and $n \geq 3$, then any bounded solution of $-\Delta u = f$ in $\mathbb{R}^n$ has the form*

$$u(x) = \int_{\mathbb{R}^n} \Phi(x-y)f(y)dy + C \quad (4.49)$$

for some constant C .

Proof. If u' is another solution, then $u - u'$ is a constant by Liouville's theorem. ■

4.2.5 Harnack's Inequality

We write $V \subseteq\subseteq U$ to denote $V \subseteq \bar{V} \subseteq U$ and $\bar{V}$ is compact, we say V is compactly contained in U .

Theorem 16. *For each connected open set $V \subseteq\subseteq U$, there exists a positive constant C depending only on V such that*

$$\sup_V u \leq C \inf_V u \quad (4.50)$$

for all non-negative harmonic functions u in U . In particular

$$\frac{1}{C}u(y) \leq u(x) \leq Cu(y) \quad (4.51)$$

for all $x, y \in V$.

Proof. Consider a ball $B(a, r) \subseteq U$, and let $x, y \in B(a, \frac{1}{4}r)$, we first claim that $B(x, \frac{1}{4}r) \subseteq B(y, \frac{3}{4}r) \subseteq B(a, r)$. Then we use mean value property and we have

$$u(x) = \int_{B(x, \frac{1}{4}r)} u(z)dz \leq \int_{B(\frac{3}{4}r, y)} u(z)dz = 3^n u(y). \quad (4.52)$$

Now since V is compact so there exists a finite cover of balls $\{B_i\}_{i=1}^M$ where $B_i \equiv B(a_i, r_i) \subseteq U$. Then for any $x, y \in V$, we may find an ordered subcover $B_{i_1}, \dots, B_{i_\ell}$ such that $x \in B_{i_1}$, $y \in B_{i_\ell}$ and $B_{i_j} \cap B_{i_{j+1}} \neq \emptyset$, and the previous conclusion holds. Then we have a chain of inequalities:

$$u(x) \leq 3^n u(x_1) \leq \dots \leq 3^{n\ell} u(y) \leq 3^{nM} u(y) \quad (4.53)$$

for all $x, y \in V$, and hence we have

$$\sup_V u \leq C \inf_V u \quad (4.54)$$

where C is a constant only depend on V . ■

4.2.6 Convergence Theorems

In this section we shall discuss some convergence theorems for harmonic functions:

Theorem 17. (*Weierstrass Convergence Theorem*) *If a sequence of harmonic functions $\{h_n\}$ converges locally uniformly on U , then the limit $h = \lim_{n \rightarrow \infty} h_n$ is also harmonic on U , furthermore the derivatives converge locally uniformly as well, i.e $\lim_{n \rightarrow \infty} D_{x_i} h_n = D_{x_i} h$.*

Proof. Let $K \subseteq U$ be compact, we choose a larger compact set $\tilde{K}$ and $r := \text{dist}(K, \mathbb{R}^n / \tilde{K}) > 0$, we use the derivative estimate to get

$$|D^\alpha h_l(a) - D^\alpha h_k(a)| \leq \frac{C}{r^{|\alpha|}} \sup_{\tilde{K}} |h_l - h_k| \quad (4.55)$$

and from the uniform convergence of $\{h_n\}$ on $\tilde{K}$, and we see that $\{D^\alpha h_n\}$ is a Cauchy sequence on K , thus uniformly convergent on K , and locally uniformly convergent on U . We can also show that $\lim_{n \rightarrow \infty} D^\alpha h_n = D^\alpha h$ locally uniformly on U for all index α . In particular we have $\Delta h = \lim_{n \rightarrow \infty} \Delta h_n$ and hence h is harmonic. ■

Theorem 18. (*Harnack Convergence Theorem*) *Consider an increasing sequence of harmonic functions $\{h_n\}_{n=1}^\infty$ on $U \subseteq \mathbb{R}^n$, then either $\lim_{n \rightarrow \infty} h_n = \infty$ or it converge locally uniformly to a harmonic function. In particular, if $\exists x_0 \in U$ such that $\lim_{n \rightarrow \infty} h_n(x_0) \neq \infty$, then we may conclude locally uniform convergence.*

Proof. Assume $x_0 \in U$ such that $\lim_{n \rightarrow \infty} h_n(x_0) \neq \infty$. For all $k \leq l$, K compact, and $x_0 \in K \in U$, we have $h_l - h_k$ as a non-negative harmonic function, and hence by Harnack's inequality,

$$\max_K (h_l - h_k) \leq C \min_K (h_l - h_k) \leq C(h_l(x_0) - h_k(x_0)) \quad (4.56)$$

which implies the uniform Cauchy property of $\{h_n\}$ on K , and hence locally uniform convergence is achieved. We now let $\lim_{n \rightarrow \infty} h_n = h$ and it remains to show h is harmonic, which follows from Weierstrass convergence theorem. ■

4.3 Green's Function

Assume $U \subseteq \mathbb{R}^n$ open and bounded, in this section, we will derive a general representation formula for Poisson's equation subject to the boundary condition

$$\begin{cases} -\Delta u = f & \text{in } U \\ u = g & \text{on } \partial U \end{cases} \quad (4.57)$$

Recall the Gauss-Green theorem introduced in the previous section, we will derive a corollary of Gauss-Green theorem:

Corollary 1. *Let $u, v \in C^2(\bar{U})$, then:*

$$(i) \int_U \Delta u dx = \int_{\partial U} \frac{\partial u}{\partial \nu} dS$$

$$(ii) \int_U Dv \cdot Dv dx = - \int_U u \Delta v dx + \int_{\partial U} \frac{\partial v}{\partial \nu} u dS$$

$$(iii) \int_U u \Delta v - v \Delta u dx = \int_{\partial U} u \frac{\partial v}{\partial \nu} - v \frac{\partial u}{\partial \nu} dS$$

where ν is the outward pointing normal unit vector field on ∂U .

Proof. For (i), recall the integration by parts formula states that

$$\int_U u'v = - \int_U uv' + \int_{\partial U} uv \cdot \nu dS \quad (4.58)$$

by letting $v \equiv 1$, then we will have (i). As for (ii), (iii), let $v = Du$. ■

Let $x \in U$, $B(x, \varepsilon) \subseteq U$, define $V_\varepsilon = U/B(x, \varepsilon)$, then by corollary (iii) and the fact that $\Delta \Phi(x-y) = 0$, $x \neq y$, we have

$$\int_{V_\varepsilon} \Phi(y-x) \Delta u(y) dy = \int_{\partial V_\varepsilon} \Phi(y-x) \frac{\partial u}{\partial \nu} - u(y) \frac{\partial \Phi}{\partial \nu} dS(y). \quad (4.59)$$

From the proof of the fundamental solution to Poisson's equation, we claimed that

$$u(x) = \lim_{\varepsilon \rightarrow 0} - \int_{\partial B(x, \varepsilon)} \frac{\partial \Phi}{\partial \nu} (y-x) u(y) dS(y) \quad (4.60)$$

where ν is the outward normal unit vector on $\partial B(x, \varepsilon)$, (the minus sign is because in Poisson's equation, ν is the inward normal), also

$$\begin{aligned} \int_{\partial V_\varepsilon} \frac{\partial \Phi}{\partial \nu} (y-x) u(y) dS(y) &= \int_{\partial U} \frac{\partial \Phi}{\partial \nu} (y-x) u(y) dS(y) - \int_{\partial B(x, \varepsilon)} \frac{\partial \Phi}{\partial \nu} (y-x) u(y) dS(y) \\ &= \int_{\partial U} \frac{\partial \Phi}{\partial \nu} (y-x) u(y) dS(y) + u(x) \end{aligned}$$

when $\varepsilon \rightarrow 0$. Similarly,

$$\int_{\partial V_\varepsilon} \Phi(y-x) \frac{\partial u}{\partial \nu} dS(y) = \int_{\partial U} \Phi(y-x) \frac{\partial u}{\partial \nu} dS(y) - \int_{\partial B(x, \varepsilon)} \Phi(y-x) \frac{\partial u}{\partial \nu} dS(y)$$

and

$$\left| \int_{\partial B(x, \varepsilon)} \Phi(y-x) \frac{\partial u}{\partial \nu} dS(y) \right| \leq C \|u\|_{L^\infty(\partial B(x, \varepsilon))} \cdot \left| \int_{\partial B(x, \varepsilon)} \Phi(y-x) dS(y) \right| \xrightarrow{\varepsilon \rightarrow 0} 0 \quad (4.61)$$

as we have shown previously. So combine all the result above and set $\varepsilon \rightarrow 0$ we have

$$\int_U \Phi(y-x) \Delta u(y) dy = \int_{\partial U} \Phi(y-x) \frac{\partial u}{\partial \nu} dS(y) - \int_{\partial U} \frac{\partial \Phi}{\partial \nu} (y-x) u(y) dS(y) - u(x)$$

which means we have

$$u(x) = \int_{\partial U} \Phi(y-x) \frac{\partial u}{\partial \nu} dS(y) - \int_{\partial U} \frac{\partial \Phi}{\partial \nu} (y-x) u(y) dS(y) - \int_U \Phi(y-x) \Delta u(y) dy \quad (4.62)$$

for any point $x \in U$ and any function $u \in C^2(\bar{U})$. So now we would be able to solve u directly if we already know the value of Δu in U and $u, \frac{\partial u}{\partial \nu}$ on ∂U since Φ is totally known to us. We can actually further rewrite the above term. For a fixed x , we introduce the following corrector function $\varphi^x = \varphi^x(y)$ defined by

$$\begin{cases} \Delta \varphi^x = 0 & \text{in } U \\ \varphi^x = \Phi(y - x) & \text{on } \partial U \end{cases} \quad (4.63)$$

Since $\Delta \varphi^x(y) = 0$ so now from (iii) of the above corollary yields

$$\begin{aligned} \int_U \varphi^x(y) \Delta u(y) dy &= \int_{\partial U} \varphi^x(y) \frac{\partial u(y)}{\partial \nu} - u(y) \frac{\partial \varphi^x(y)}{\partial \nu} dS(y) \\ &= \int_{\partial U} \Phi(y - x) \frac{\partial u(y)}{\partial \nu} - u(y) \frac{\partial \varphi^x(y)}{\partial \nu} dS(y). \end{aligned}$$

This motivates the definition of Green's function

Definition 18. *The Green's function for the region U is*

$$G(x, y) := \Phi(y - x) - \varphi^x(y) \quad (4.64)$$

for all $x \neq y; x, y \in U$.

Then we have

$$u(x) = - \int_{\partial U} u(y) \frac{\partial G}{\partial \nu}(x, y) dS(y) - \int_U G(x, y) \Delta u(y) dy \quad (4.65)$$

where

$$\frac{\partial G}{\partial \nu}(x, y) = D_y G(x, y) \nu(y) \quad (4.66)$$

is the outer normal derivative of G w.r.t y . Here the nice thing is that we no longer have $\frac{\partial u}{\partial \nu}$. If we use the condition on u such that $-\Delta u = f$ in U and $u = g$ in ∂U , we would have the representation formula using Green's function:

Theorem 19. *If $u \in C^2(\bar{U})$ solves the problem $\begin{cases} -\Delta u = f & \text{in } U \\ u = g & \text{on } \partial U \end{cases}$, then*

$$u(x) = - \int_{\partial U} g(y) \frac{\partial G}{\partial \nu}(x, y) dS(y) + \int_U f(y) G(x, y) dy \quad (4.67)$$

for all $x \in U$.

Theorem 20. *Green's function is symmetric, i.e for all $x \neq y; x, y \in U$, $G(x, y) = G(y, x)$.*

Proof. Fix $x, y \in U, x \neq y$, define $v(z) = G(x, z), w(z) = G(y, z)$ where $z \in U$. Then by definition $v(z), w(z)$ are both harmonic hence $\Delta w(z) = \Delta v(z) = 0$ when $x \neq z; y \neq z$, and also $w(z) = v(z) = 0$ on ∂U . A typical confusion is that some might here use maximal principle to conclude $w(z) = 0, v(z) = 0$ on U , but note that we have $-\Delta w(z) = f, -\Delta v(z) = f$ are not harmonic! We now apply Green's identity on $V = U - B(x, \varepsilon) \cup B(y, \varepsilon)$ and hence by (iii) of the above corollary we have

$$\int_{\partial B(x, \varepsilon)} \frac{\partial v}{\partial \nu} w - \frac{\partial w}{\partial \nu} v dS(z) = \int_{\partial B(y, \varepsilon)} \frac{\partial w}{\partial \nu} v - \frac{\partial v}{\partial \nu} w dS(z) \quad (4.68)$$

where ν is the inward pointing unit normal vector field on $\partial B(x, \varepsilon) \cup \partial B(y, \varepsilon)$. Now

$$\left| \int_{\partial B(x, \varepsilon)} \frac{\partial w}{\partial \nu} v dS \right| \leq C \|w\|_{L^\infty(\partial B(x, \varepsilon))} \cdot \left| \int_{\partial B(x, \varepsilon)} v dS \right| \xrightarrow{\varepsilon \rightarrow 0} 0, \quad (4.69)$$

■

4.4 Green's Function for Half-Space

Consider the half space in $\mathbb{R}^n$ defined by

$$\mathbb{R}_+^n = \{x = (x_1, \dots, x_n) \in \mathbb{R}^n, x_i > 0\} \quad (4.70)$$

we say the *reflection* of x in the plane $\partial \mathbb{R}_+^n$ is the point

$$\tilde{x} = (x_1, \dots, x_{n-1}, -x_n). \quad (4.71)$$

The goal is to build the Green's function for this specific type of region, and we define the corrector function as

$$\varphi^x(y) = \begin{cases} \Phi(y - \tilde{x}) & y \in \mathbb{R}_+^n \\ \Phi(y - x) & y \in \partial \mathbb{R}_+^n \end{cases} \quad (4.72)$$

and we have the following equation:

$$\begin{cases} -\Delta \varphi(y) = 0 & \text{in } \mathbb{R}_+^n \\ \varphi^x(y) = \Phi(y - x) & \text{on } \partial \mathbb{R}_+^n \end{cases}. \quad (4.73)$$

Definition 19. *The Green's function for the half space $\mathbb{R}_+^n$ is*

$$G(x, y) = \Phi(y - x) - \Phi(y - \tilde{x}) \quad (4.74)$$

for $x \neq y; x, y \in \mathbb{R}_+^n$.

Now we have

$$\frac{\partial G}{\partial y_n} = \frac{\partial \Phi}{\partial y_n}(y - x) - \frac{\partial \Phi}{\partial y_n}(y - \tilde{x}) = -\frac{1}{n\alpha(n)} \left(\frac{y_n - x_n}{|y - x|^n} - \frac{y_n + x_n}{|y - \tilde{x}|^n} \right) \quad (4.75)$$

if $y \in \partial \mathbb{R}_+^n$, then

$$\frac{\partial G}{\partial \nu}(x, y) = -\frac{\partial G}{\partial y_n}(x, y) = \frac{2x_n}{n\alpha(n)} \frac{1}{|x - y|^n}. \quad (4.76)$$

Now suppose u solves the boundary value problem $\begin{cases} \Delta u = 0 & \text{in } \mathbb{R}_+^n \\ u = g & \text{on } \partial \mathbb{R}_+^n \end{cases}$, then use the representation formula of Green's function, we will have

$$u(x) = \frac{2x_n}{n\alpha(n)} \int_{\partial \mathbb{R}_+^n} \frac{g(y)}{|x - y|^n} dy \quad (4.77)$$

for $x \in \mathbb{R}_+^n$.

Definition 20. *The function*

$$K(x, y) = \frac{2x_n}{n\alpha(n)} \frac{1}{|x - y|^n} \quad (4.78)$$

where $x \in \mathbb{R}_+^n, y \in \partial\mathbb{R}_+^n$ is called the Poisson's kernel for $\mathbb{R}_+^n$, and

$$u(x) = \frac{2x_n}{n\alpha(n)} \int_{\partial\mathbb{R}_+^n} \frac{g(y)}{|x - y|^n} dy \quad (4.79)$$

is Poisson's formula.

Theorem 21. *(Poisson's formula for half-space) Assume $g \in C(\mathbb{R}^{n-1}) \cap L^\infty(\mathbb{R}^{n-1})$ and define*

$$u(x) = \frac{2x_n}{n\alpha(n)} \int_{\partial\mathbb{R}_+^n} \frac{g(y)}{|x - y|^n} dy, x \in \mathbb{R}_+^n \quad (4.80)$$

then the followings hold:

(i) $u \in C^\infty(\mathbb{R}_+^n) \cap L^\infty(\mathbb{R}_+^n)$;

(ii) $\Delta u = 0$ in $\mathbb{R}_+^n$ and $\lim_{x \rightarrow x^0} u(x) = g(x^0)$ for each point $x^0 \in \partial\mathbb{R}_+^n$ and $x \in \mathbb{R}_+^n$.

Proof. For (i), for each fixed x , we know that the map $y \mapsto G(x, y)$ is harmonic except for $y = x$, also $G(x, y) = G(y, x)$, so the map $x \mapsto G(x, y)$ is harmonic except for $x = y$, hence $x \mapsto -\frac{\partial G}{\partial y_n}(x, y) = K(x, y)$ is harmonic for $x \in \mathbb{R}_+^n, y \in \partial\mathbb{R}_+^n$. ■

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